



## OBJECTIVE

Final-year MSc student at ISAE-SUPAERO seeking a 6-month end-of-studies internship from April 2027. Interested in flight control, guidance and navigation, and in bridging aerospace engineering with software to develop autonomous and reliable systems.

## TECHNICAL SKILLS

### CONTROL & GNC

Flight Dynamics, Flight Control, Guidance & Navigation, Adaptive Control, PID/MRAC, 6-DOF Dynamics

### PROGRAMMING

Python, C, MATLAB/Simulink, Machine Learning, ROS/Gazebo, TLA+, Git/SVN

### SYSTEMS & TOOLS

Systems Engineering, Modelling & Simulation, CATIA, CFD, STK, Ansys HFSS

## CERTIFICATION

Systems Tool Kit (STK)- Level 1 (Ansys)

EDX – 16.00x: Introduction to Aerospace Engineering: Astronautics and Human Space Flight (Massachusetts Institute of Technology)

## SOFT SKILLS

Leadership  
Professional Responsibility  
Analytical Thinking  
Adaptability

## LANGUAGES

|         |                   |
|---------|-------------------|
| English | Native (C2)       |
| French  | Intermediate (B1) |
| Hindi   | Native (C2)       |

## ACTIVITIES

### Vice President - Outgoing Global Exchange at AIESEC Carleton

Lead a team of exchange managers facilitating exchanges for internship and volunteer opportunities for participants.

# ANUSHA JAIN

ISAE-SUPAERO - MSc Aerospace Engineering | Systems & Control  
Flight Control • Guidance & Navigation • Aerospace Software

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## EXPERIENCE



### ISAE-SUPAERO – Graduate Student Researcher | Glider Control & RL

Feb'26 - Present  
Toulouse, France

Atmosphere-Aware RL Controller for High-Altitude Glider Flight

- Developing a 6DOF glider simulator in MATLAB/Simulink with atmospheric turbulence models.
- Designing an adaptive RL controller, benchmarked against a PID baseline (in progress).



### SupaeroMoon

Sep'25 - Present  
Toulouse, France

Head of Systems Engineering

- Leading systems engineering across three rover projects (ERC 2026, MDRS, open-source modular rover).
- Defining architecture and interfaces between autonomy, structures and science subsystems; tracking compliance with mission requirements.



### BlackBerry QNX – CO-OP

May'23 - Aug'23  
Ottawa, Canada

Safety Systems Software Developer

- Drafted and revised safety manuals for Automotive OEM and RTOS products.
- Automated code traceability (Python, C, JSON) and supported HARA/HAZOP safety assessments using TLA+ formal modelling.

## ACADEMIC PROJECTS

### UAV Flight Control – Adaptive Control

Apr'26 - Jun'26

- Designed a cascade PID baseline, then implemented a 3-axis MRAC (MIT rule) altitude controller for a quadrotor under sudden mass drop.
- Validated in Simulink simulation and physical flight tests.

### Airbus Fly Your Ideas Challenge

Mar'26 - Jun'26

- Led a multidisciplinary team to Round 2 applying post-quantum cryptography to aerospace cybersecurity.
- Directed technical development, system design, and team coordination with mentors and teammates.

### Cooperative UAV Simulation & Control – ROS/Gazebo

Sep'24 - Apr'25

- Integrated and simulated a dual-drone cooperative payload system using ROS, MAVROS, ArduCopter, and Gazebo.
- Led project integration, aligning system architecture, control design, and simulation.

### Spacecraft Attitude Dynamics and Control

Sep'24 - Dec'24

- Modelled spacecraft attitude dynamics and reaction-wheel momentum exchange in MATLAB/Simulink with quaternion feedback control.
- Built an orbit propagator and modelled four perturbation torques (SRP, drag, gravity gradient, magnetic) to validate attitude response.

## EDUCATION



MSc in Aerospace Engineering, Aerospace Systems and Control

2025 - 2027

Toulouse, France

### ISAE SUPAERO

- Space Propulsion, Adaptive Control, Artificial Intelligence for Control, Multiple-Input Multiple- Output (MIMO) Systems, Hybrid Control, Robust and Optimal Control.



B.Eng. in Aerospace, Space Systems Design

2021 - 2025

Ottawa, Canada

### Carleton University

- Transatmospheric Space Propulsion, Orbital Mechanics, Spacecraft Attitude Dynamics and Control, Spacecraft Design, Heat Transfer for Aerospace Applications.